

ARTIFICIAL INTELLIGENCE REPLACE ETHICAL HAKER AND CYBERSECURITY JOBS

Tanuja Shankar Chikane
Student, Dept. of CS,
ICLES' Motilal Jhunjhunwala College,
Mumbai University,
Navi Mumbai, India
tanujachikane2005@gmail.com

Miss. Shruti Baikar
Assistant Professor, Dept. of IT & CS,
ICLES' Motilal Jhunjhunwala College,
Mumbai University,
Navi Mumbai, India
shrutibaikaricles@gmail.com

Mrs. Priyanka Jadhav
HOD, Dept of IT & CS,
ICLES' Motilal Jhunjhunwala College,
Mumbai University,
Navi Mumbai, India
manepri28@gmail.com

Abstract :

AI has the capacity to segregate vulnerabilities and finding all the staging data and analyzing them but it can't bring the expertise that humans have. We need humans for accountability. A software cannot take accountability and hence cyber security professional Will always be in demand. Overall, we conclude that AI can only boost not replace human expertise in the fast-evolving threat landscape.

Keywords :

Artificial Intelligence (AI), Cybersecurity Automation, Employment, AI in Cybersecurity, Human-AI Collaboration in Cybersecurity, Automation vs. Human Expertise, job opportunities and challenges, Future of Cybersecurity Jobs.

INTRODUCTION :

The 21st century has witnessed an unprecedented coverage of technological advancements, revolutionizing the way we live, work and interact with each other. Artificial Intelligence (AI) is reshaping the cybersecurity landscape by improving threat detection, incident response, and risk assessment. AI-powered tools, including automated security systems, machine learning-based anomaly detection, and predictive analytics, enable faster identification and mitigation of cyber threats compared to traditional security approaches. These advancements

have sparked discussions about whether AI could eventually replace human cybersecurity professionals.

While AI is highly effective at analyzing large datasets, identifying patterns, and detecting security breaches in real time, cybersecurity remains a complex and ever-evolving field that requires human expertise. Cybercriminals continuously develop new attack methods, such as zero-day vulnerabilities and AI-generated malware, which AI systems may struggle to recognize. Furthermore, **adversarial attacks** can be used to manipulate AI algorithms, misleading them into failing to detect threats.

Rather than replacing cybersecurity professionals, AI is expected to enhance their capabilities by handling repetitive security operations, allowing experts to focus on advanced threat analysis, decision-making, and risk mitigation. The future of cybersecurity will depend on **collaboration between AI and human intelligence**, ensuring a more **adaptive, efficient, and resilient defense** against emerging cyber threats.

Problem Definition :

The growing use of Artificial Intelligence (AI) in cybersecurity has led to debates about whether it will replace human professionals. AI enhances threat detection, risk assessment, and incident response, allowing organizations to analyze large datasets and identify cyber threats in real time. However, cybersecurity is dynamic, requiring

human expertise to handle evolving threats, ethical concerns, and strategic decision-making—areas where AI has limitations. Rather than replacing cybersecurity professionals, AI is best suited to complement them, automating routine tasks while experts focus on advanced threat mitigation, policy enforcement, and ethical security practices. A collaborative approach between AI and human intelligence is key to building a resilient cybersecurity framework.

The challenge is not whether AI will replace cybersecurity professionals but how both can collaborate to enhance security measures. A balanced approach is essential, where AI improves threat detection efficiency, while human experts contribute strategic planning, ethical oversight, and advanced threat response. To tackle this issue effectively, organizations must invest in AI advancements, workforce training, and policies that promote a cooperative cybersecurity framework, ensuring AI serves as an enhancement rather than a complete replacement for human expertise.

Research Methodology :

Artificial Intelligence (AI) will replace cybersecurity jobs in the future through a structured qualitative and quantitative approach. The study will use descriptive and analytical research methods to evaluate AI's effectiveness in threat detection, automation, and workforce transformation in cybersecurity. Data will be collected from primary sources, such as surveys and interviews with cybersecurity professionals, AI developers, and IT experts, along with case studies of organizations implementing AI-driven security solutions. Additionally, secondary data from academic research, industry reports,

and government policies will be analysed to understand AI's impact on cybersecurity employment. The research will incorporate quantitative analysis to assess trends in job displacement, AI adoption rates, and workforce demand, while qualitative analysis will examine expert insights, ethical concerns, and real-world AI applications. The evaluation will focus on AI's capabilities, limitations, automation potential, and ethical implications, determining whether AI will replace, assist, or transform cybersecurity roles. Ultimately, the study will provide insights to help organizations, policymakers, and professionals prepare for an AI-integrated cybersecurity landscape that emphasizes collaboration rather than full automation.

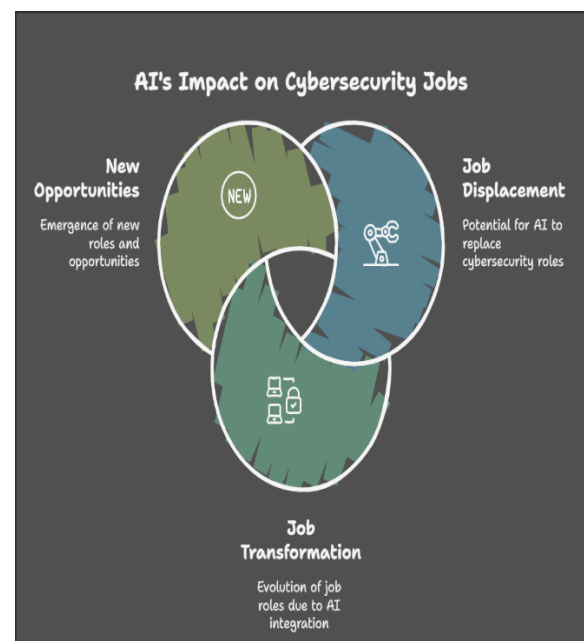


Fig.1 AI's Impact On Cybersecurity Jobs

The image presents a Venn diagram that highlights the impact of AI on cybersecurity jobs, focusing on three key areas:

1. Emerging Opportunities – AI is creating new roles within cybersecurity, such as AI-powered threat analysis and automated security operations.
2. Job Displacement – There is a risk that AI may replace certain tasks,

particularly those that involve routine or automated cybersecurity processes.

3. Role Evolution – Many cybersecurity jobs will adapt and evolve with AI integration, requiring professionals to enhance their technical and analytical skills.

The interconnected nature of these aspects suggests that AI will transform the cybersecurity workforce rather than fully replace professionals. The future of cybersecurity jobs will likely involve AI complementing human expertise, leading to a more efficient and collaborative cybersecurity environment.

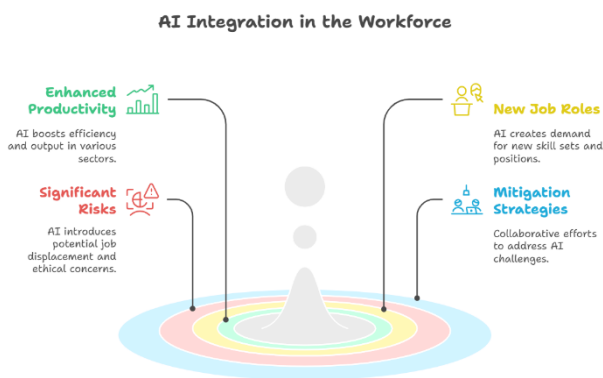


Fig.2 AI integration in the workplace

The image showcases the role of AI in the workforce, emphasizing its advantages and challenges. It highlights four main aspects: Increased Efficiency, Potential Risks, Emerging Job Opportunities, and Approaches to Mitigation. AI improves productivity across industries while also posing risks such as job loss and ethical dilemmas. Additionally, it leads to the creation of new job roles that demand different skills. The image also stresses the importance of collaborative strategies to manage AI-related challenges, presenting a balanced perspective on its impact on employment.

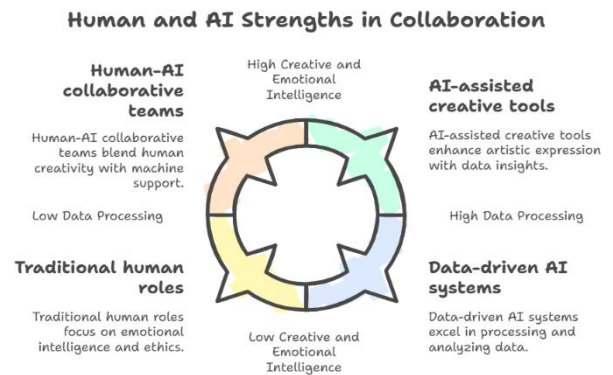


Fig.3. Human and AI strength in collaboration

The image presents a framework for human and AI collaboration, highlighting their respective strengths and weaknesses.

1. Human-AI Collaborative Teams

- Merge human creativity with AI efficiency.
- Strong in creativity and emotional intelligence, but weak in data processing.

2. AI-Assisted Creative Tools

- Support artistic work with AI-driven insights.
- Excel in data processing, but lack creativity.

3. Traditional Human Roles

- Prioritize emotional intelligence and ethical decision-making.
- Strong in creativity and ethics, but poor at handling large data.

4. Data-Driven AI Systems

- Specialize in analyzing and processing data efficiently.
- Strong in data analysis, but weak in creativity and emotional intelligence.

Humans and AI bring unique abilities to the table. While AI thrives in data processing and emotional intelligence.

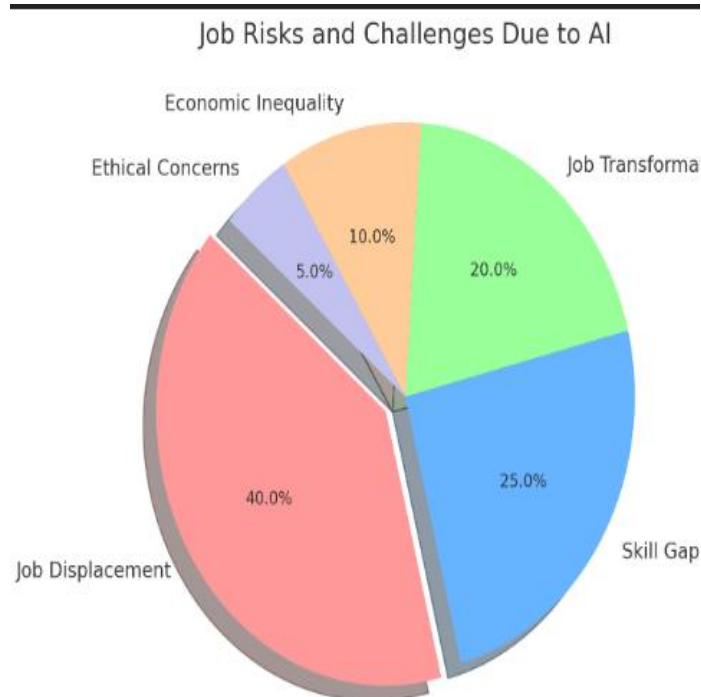
Literature survey:

Sr No	Paper	Author	Advantages	Drawbacks
1	Will Artificial Intelligence take over our jobs a human Perspective on Employment future	Bhajneet Kaur, Sanskriti Sharma	Discusses how replace certain jobs, but also create new ones focuses on AI's impact on jobs considering the social and economic implications also examine positive as well as negative impacts of AI on employment.	Focuses on identifying the challenges and implementation of AI's impact on employment, without offering detailed concrete solutions. Paper was published in 2016 the data and statistics may not reflect the todays state of AI's impact on employment
2	The Impact of artificial intelligence on employment	Georgios Petropoulos	As published in 2018 where impact of AI on employment was becoming increasingly relevant where as author discusses policy implications and potential solutions to mitigate the negative effects of AI on employment afterall paper is well organised,making it easy to follow the author's reasoning and arguments	Limited discussion of AI 's potential benefits. As the field of AI and employment is rapidly growing, the paper's findings and conclusions may require updates to reflect the latest developments
3	Let The Robots Take Over: How The Future Of AI Will Create More Jobs	Kenneth Coats	Provides a concrete examples of industries and companies that have created new jobs due to AI adoption,focuses on human-AI collaboration to create new job opportunities	No clear policy recommendations to support the creation of new jobs in an AI driven economy lack of empirical evidence or data
4	Millions of Americans have lost jobs in the pandemic— And robots and AI are replacing them faster than ever	Alana Semuels	The article the give a talk to impact of COVID-19 pandemic on employment. author uses clear and concise language making the article easy to understand for readers also provides examples of companies and industries they have implemented automation technologies	Provides a general overview but does not offer a detailed impact of automation on employment. the article depends on report and studies from other organizations, rather than offering original data

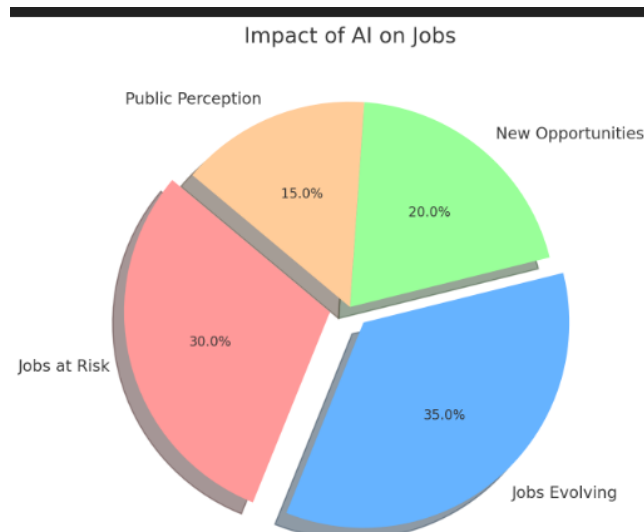
Table1

The above literature survey table provides details on various research papers that address the opportunities challenges and implementations on employments.

Study survey:



Here is a pie chart illustrating the various job risks and challenges posed by AI. The chart highlights that job displacement is the most significant concern (40%), followed by the skill gap (25%), job transformation (20%), economic inequality (10%), and ethical concerns (5%). The largest section, job displacement, emphasizes the risk of automation replacing repetitive jobs, while other factors show the need for adaptation and fairness in the evolving workforce. Let me know if you need any modifications!



AI is expected to impact jobs in various ways. Around 30% of routine-based roles,

such as data entry and telemarketing, are at risk of automation. However, 35% of jobs will evolve, integrating AI as an assistant rather than a replacement, particularly in healthcare, education, and creative fields. Additionally, 20% of new opportunities will emerge, including AI engineering and ethics consulting. Public perception remains mixed, with 15% expressing both anxiety and optimism about AI's influence. To adapt, reskilling and AI-human collaboration will be crucial in ensuring a balanced future workforce.

Conclusion:

AI's impact on employment is more about transformation than outright job loss. While some routine tasks will be automated, AI will also introduce new job opportunities and improve existing roles through collaboration. Many professions will adapt, requiring workers to upskill in areas where human creativity, emotional intelligence, and ethical judgment are vital. Rather than entirely replacing jobs, AI will reshape the workforce, emphasizing the importance of continuous learning and flexibility. The future of work depends on integrating AI as a tool to enhance productivity rather than perceiving it as a threat to employment.

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2. Martin Ford (2015) – "Rise of the Robots" Explores AI's potential to cause mass unemployment.
3. Brynjolfsson & McAfee (2011) – "Race Against the Machine" Discusses AI's role in productivity and job displacement.
4. Kinder et al. (2023) – "Generative AI and the Future of Work" Examines AI's impact on U.S. occupations.
5. Najana & Chintale (2022) – "AI's Impact on Employment Patterns" Highlights AI's risks and opportunities for workers.
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